GIA - IAA
Introducció a l’Aprenentatge Automàtic

9. Lab: Itemsets. Apriori Algorithm

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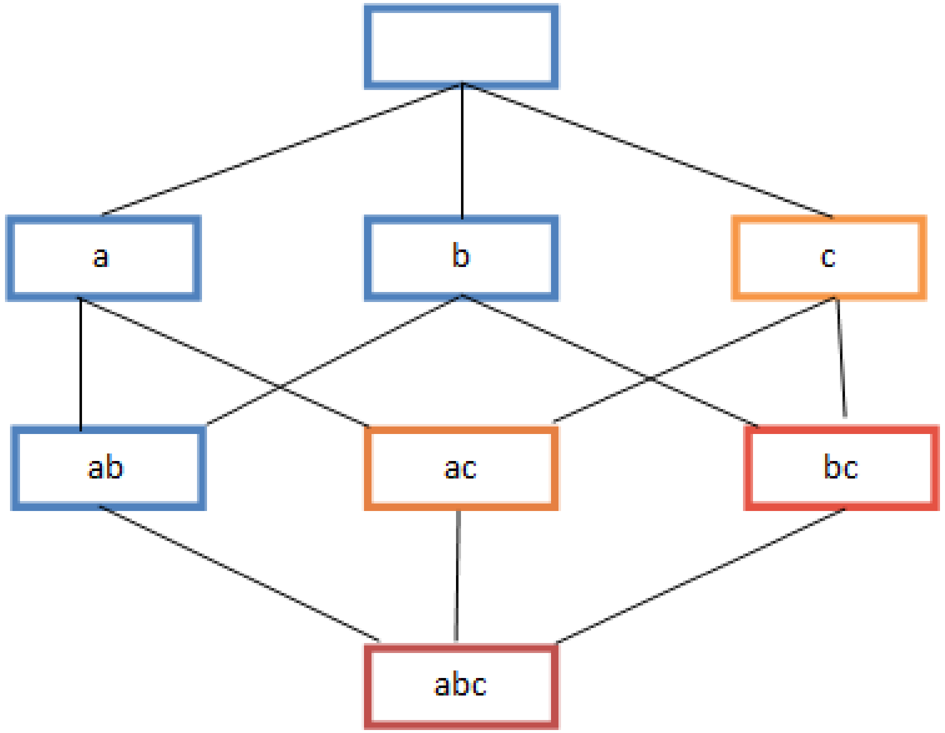
Association Rule Learning

- Frequent itemset lattice, where the color of the box indicates how many transactions contain combination of items. Note that lower levels of the lattice can contain at most the minimum number of their parents' items ; e.g. {ac} can have only at most items. This is called the *downward-closure.*



https://en.wikipedia.org/wiki/Association_rule_learning

Metrics

```python
dataset = [['Milk', 'Onion', 'Nutmeg', 'Kidney Beans', 'Eggs', 'Yogurt'],
           ['Dill', 'Onion', 'Nutmeg', 'Kidney Beans', 'Eggs', 'Yogurt'],
           ['Milk', 'Apple', 'Kidney Beans', 'Eggs'],
           ['Milk', 'Unicorn', 'Corn', 'Kidney Beans', 'Yogurt'],
           ['Corn', 'Onion', 'Onion', 'Kidney Beans', 'Ice cream', 'Eggs']]

```

$$Support = \frac{frq(X,Y)}{N}$$

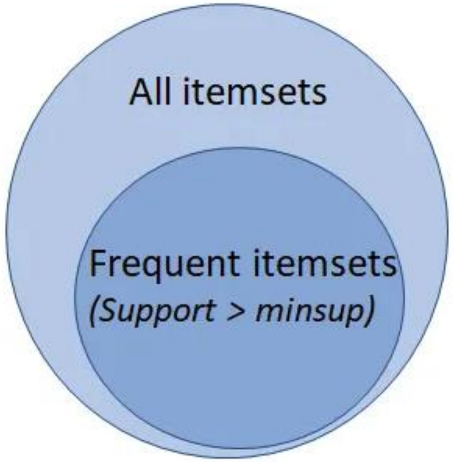
$$Rule:\ X \Rightarrow Y$$

$$Confidence = \frac{frq(X,Y)}{frq(X)}$$

$$Lift = \frac{Support}{Supp(X) \times Supp(Y)}$$

Apriori Algorithm

- Apriori is a popular algorithm [1] for extracting frequent itemsets with applications in association rule learning. The apriori algorithm has been designed to operate on databases containing transactions, such as purchases by customers of a store. An itemset is considered as "frequent" if it meets a user-specified support threshold. For instance, if the support threshold is set to 0.5 (50%), a frequent itemset is defined as a set of items that occur together in at least 50% of all transactions in the database.



[1] Agrawal, Rakesh, and Ramakrishnan Srikant. "Fast algorithms for mining association rules." Proc. 20th int. conf. very large data bases, VLDB. Vol. 1215. 1994.

mlxtend library

```python
from mlxtend.frequent_patterns import apriori

frequent_itemsets = apriori(df, min_support=0.6, use_colnames=True)
frequent_itemsets['length'] = frequent_itemsets['itemsets'].apply(lambda x: len(x))
frequent_itemsets
```

[131] ✓ 0.0s

	support	itemsets	length
0	0.800	(Eggs)	1
1	1.000	(Kidney Beans)	1
2	0.600	(Milk)	1
3	0.600	(Onion)	1
4	0.600	(Yogurt)	1

```python
from mlxtend.frequent_patterns import association_rules

association_rules(frequent_itemsets, metric="confidence", min_threshold=0.7)
```

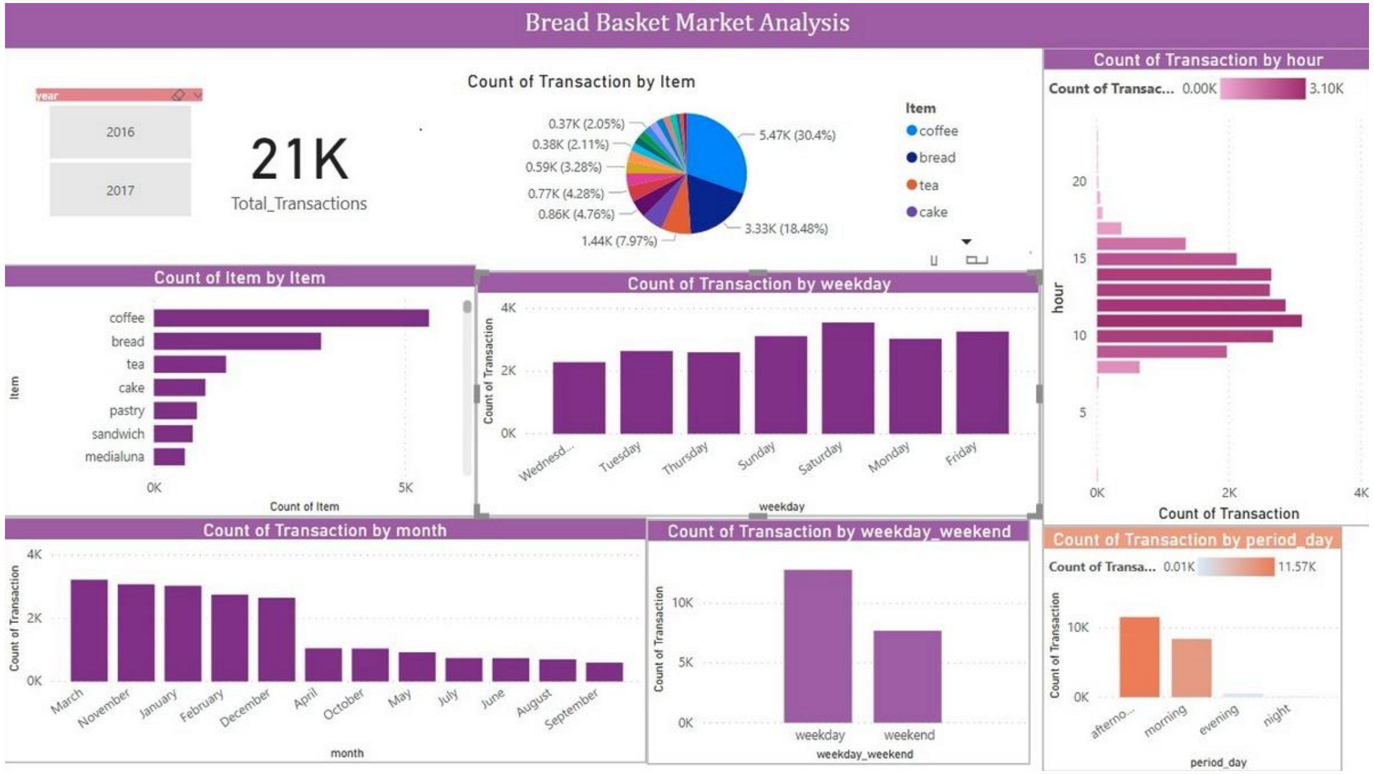
[137] ✓ 0.0s

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	leverage	conviction	zhangs_metric
0	(Eggs)	(Kidney Beans)	0.800	1.000	0.800	1.000	1.000	0.000	inf	0.000
1	(Kidney Beans)	(Eggs)	1.000	0.800	0.800	0.800	1.000	0.000	1.000	0.000
2	(Onion)	(Eggs)	0.600	0.800	0.600	1.000	1.250	0.120	inf	0.500
3	(Eggs)	(Onion)	0.800	0.600	0.600	0.750	1.250	0.120	1.600	1.000
4	(Milk)	(Kidney Beans)	0.600	1.000	0.600	1.000	1.000	0.000	inf	0.000
5	(Onion)	(Kidney Beans)	0.600	1.000	0.600	1.000	1.000	0.000	inf	0.000

https://rasbt.github.io/mlxtend/user_guide/frequent_patterns/apriori/

Bread Basket Dataset

- This Kaggle dataset belongs to "The Bread Basket" a bakery located in Edinburgh. The dataset has around 31K entries, over 21K transactions, and 4 columns. The dataset has transactions of customers who ordered different items from this bakery online and the time period of the data is from 26-01-03 to 27-12-03.



Groceries Dataset

PLAYTIME: Analyse the Groceries dataset and find the most frequent itemsets and association rules.

```
basket = pd.read_csv('../../data/Groceries_dataset.csv')
basket
```

[195] ✓ 0.0s

	Member_number	Date	itemDescription
0	1808	21-07-2015	tropical fruit
1	2552	05-01-2015	whole milk
2	2300	19-09-2015	pip fruit
3	1187	12-12-2015	other vegetables
4	3037	01-02-2015	whole milk
...	...	...	...
38760	4471	08-10-2014	sliced cheese
38761	2022	23-02-2014	candy
38762	1097	16-04-2014	cake bar
38763	1510	03-12-2014	fruit/vegetable juice
38764	1521	26-12-2014	cat food

38765 rows × 3 columns